Vacuum-Flux Decomposition and a Grid Shape Model for Magnetics-Free Equilibrium Reconstruction

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Abstract

Methods report for the NeurIPS 2026 Fusion Equilibrium Challenge, status 11 October 2026 (development phase). We reconstruct the DIII-D poloidal flux map $\psi(R, Z)$, q_{95} and β_N from coil currents and Thomson-scattering profiles. ψ is split into the analytic vacuum flux of the shipped coils and a learned plasma term, predicted by a fixed-weight blend of 47 components: residual MLPs on PCA coefficients, XGBoost trees, a shape-target MLP ensemble and three FiLM-conditioned U-Nets on the 65×65 grid. The best official DIII-D score is $S = 0.993927$ (rank 29 of 50), up from 0.990496 for the first entry; the leaders are at 0.9964–0.9967. Every blend weight since 8 October was chosen on 160 never-scored held-out shots. MAST (Challenge 2) is zero-shot and label-free: $S_{\mathrm{MAST}} = 0.627752$, $G = 0.631588$. We report the full funnel, the closed directions, and what is not established.

1 Task

Each tokamak shot gives coil currents, plasma current I_p and Thomson T_e/n_e profiles; the targets are the EFIT flux map on a 65×65 grid and q_{95}, β_N at every EFIT time. Challenge 1 scores DIII-D with

$$S = 0.55\, R^2_{\psi} + 0.15\, R^2_{q_{95},\beta_N} + 0.10\,(1 - D_{\mathrm{LCFS}}) + 0.20\,\mathrm{Consistency},$$

where Consistency is the mean R^2 of seven scalars the scorer derives from the predicted flux (R_{axis}, Z_{axis}, κ, upper and lower triangularity, volume, l_i). Challenge 2 scores zero-shot transfer to MAST as $G = S_{\mathrm{MAST}}/S_{\mathrm{DIII\text{-}D}}$. The data are 7,041 `diii_d_train` shots, 874 `diii_d_public_test` shots and 1,206 `mast_public_test` shots; there is no MAST training set, and the three labelled MAST demo shots are for inspection only (rule 3). The development phase closes 18 October; the Final phase (19–26 October, 3 submissions) requires a reproduction package that trains within 24 A100-hours.

2 Method

Inputs and vacuum flux. Coil currents and I_p are interpolated onto the EFIT times (with the organisers' I_p time-axis errata applied), and Thomson profiles are resampled onto fixed spatial stations so shots with different channel layouts share one feature space. The vacuum flux $\psi_{\mathrm{vac}}(R, Z) = \sum_c I_c G_c(R, Z)$ uses circular-filament elliptic-integral Green's functions over each shipped coil rectangle; it is exact, linear, and defined on any machine from its geometry alone (`fusioneq/greens.py`).

Base ensemble (flux level). Residual MLPs (2048 wide, depth 10, 120 epochs, bf16, a 50/50 mix of the coefficient and pixel losses) predict PCA coefficients of the plasma flux plus q_{95} and β_N. Fifteen members are averaged (v5, v6 and v8 seeds and region-loss runs, and six ramp-weighted v14 members), and XGBoost on the host network's inputs for the top 32 PCA coefficients is blended in at 0.1. The submitted q_{95} and β_N come from this ensemble.

Shape-target hybrid. 27 MLPs (v16–v21) predict the flux *shape*: ψ centred and scaled by its own mean and spread inside the scorer's envelope. Their average is moved onto the base flux's envelope mean and spread (`match_envelope`), so it contributes shape (consistency, boundary) without its weaker level. XGBoost trees on the top 32 shape-PCA coefficients (host `v21_normalize_seeds_i00`) add a second shape family.

Grid shape model. A U-Net on the grid padded to 72×72 (three downsamplings) maps the vacuum flux in shape units and the R, Z coordinate maps to the flux shape; the frame's feature vector (coil currents, I_p, Thomson with ± 4 frames of context) enters through an MLP that scales and shifts every level's channels (FiLM). The loss is pixel MSE in shape units, weighted up inside the envelope; training uses AdamW with a one-cycle schedule and bf16 autocast, keeping the epoch with the lowest validation envelope loss (`fusioneq/grid.py`, `docs/specs/2026-10-08-grid-shape-model-design.md`). Table 1 lists the trained variants.

Table 1: Grid models. Envelope loss is on the 40 never-scored validation shots.

Model	Data	Channels / epochs	Validation envelope loss
grid_v1	every 2nd frame	32 / 10	0.00105
grid_v2	all frames	48 / 15	0.00075
grid_v2_s1	all frames, seed 1	48 / 15	0.00065
grid_v3	all frames	64 / 20	0.00064
grid_v2_gs	all frames, Grad-Shafranov loss weight 50	48 / 15	0.0031 (closed)

Blend. The submitted flux is $\psi = w_b\,\text{base} + w_h\,\text{hybrid} + w_{st}\,\text{shape trees} + w_g\,\overline{\text{grid}}$, with each shape component moved onto the base's envelope statistics (`fusioneq/blend.py`). The official best (`configs/blends/v21_27_grids3.yaml`) uses $w_b = 0.10$, $w_h = 0.10$, $w_{st} = 0.15$, $w_g = 0.65$ over grid_v2, grid_v2_s1 and grid_v3. MAST uses no learned model (Section 6).

3 Evaluation protocol

A fixed shot-level split (seed 0) holds out 200 training shots. The first 40 are the *scoring* shots that every local number in the sweep writeups is reported on; the other 160 (40 *validation* plus 120 *spare*) are never used for those scores. From 8 October, every blend weight is screened on five 40-shot groups (spare thirds 1–3, validation, scoring) as paired deltas against the current official best, and pooled over the 160 never-scored shots (`docs/experiments/fusioneq/screens.yaml`). Two promotion gates (`docs/experiments/fusioneq/verdicts.yaml`):

- **G1**: beats the matched-seed baseline on the 40 scoring shots by more than seed noise (0.0013 for one model, about 0.0002 for an ensemble).
- **G2**: lifts the blend on held-out shots (from 8 October: wins most 40-shot groups of the 160 never-scored shots), then beats the official best on Codabench.

Both gates were tightened after seeing results. The seed-noise threshold in G1 was measured in v6 (seeds 1 and 2 of the v5 winner scored 0.9901–0.9902 against 0.9915), after v5 had been promoted on what turned out to be a lucky seed. The 160-shot group rule in G2 was added on 8 October after two

local gains failed to transfer: temporal smoothing (+0.00018 on the scoring shots; 0.992144 officially against 0.992168) and v21's six extra members (+0.00007 locally, -0.000004 officially). Neither gate was pre-registered for the experiments that motivated it. The MAST solenoid-shape correction (`4c428222`) was chosen by official feedback, not by a local gate.

4 Results on DIII-D

Table 2: Official DIII-D submissions that set a new best (`results/codabench.json`). D is D_{LCFS}; rank is on the Codabench public test where recorded.

Date	HF commit	S	R^2_ψ	D	Cons.	What changed
29 Sep	`15903ed1`	0.990496	0.999586	0.013063	0.963719	one residual MLP (v5), rank 30
29 Sep	`d992d040`	0.991639	0.999730	0.012340	0.968326	3-seed ensemble, rank 29
30 Sep	`07a1ba61`	0.992005	0.999758	0.012114	0.969830	5-model ensemble, rank 28
30 Sep	`df98320e`	0.992168	0.999765	0.012034	0.970494	9-model ensemble (v8), rank 26 of 77
3 Oct	`ccb9ef31`	0.992317	0.999762	0.012058	0.971258	+ 0.1 XGBoost coefficient trees
4 Oct	`f4205360`	0.992341	0.999753	0.012031	0.971439	+ six ramp-weighted members
5 Oct	`9e381a13`	0.992663	0.999759	0.011780	0.972906	+ shape-target hybrid (3 members)
6 Oct	`5cae79d2`	0.992830	0.999760	0.011645	0.973671	6 shape members at 0.5
6 Oct	`0c9a8ac9`	0.992904	0.999760	0.011559	0.973998	12 shape members at 0.65
7 Oct	`7a56d52a`	0.993078	0.999762	0.011437	0.974805	21 members + shape trees, rank 28
8 Oct	`12068a0d`	0.993114	0.999764	0.011437	0.974978	trees on 2x data, rank 28
9 Oct	`93605aab`	0.993186	0.999767	0.011510	0.975366	+ grid_v1 at 0.1, rank 29
9 Oct	`d3076dea`	0.993520	0.999773	0.011495	0.977012	grid_v2 at 0.5, rank 28
10 Oct	`9c2c5fab`	0.993769	0.999776	0.011233	0.978119	two grid seeds at 0.6, rank 28
11 Oct	`07cfd3ff`	**0.993927**	0.999777	0.011025	0.978802	three grids at 0.65, rank 29 of 50

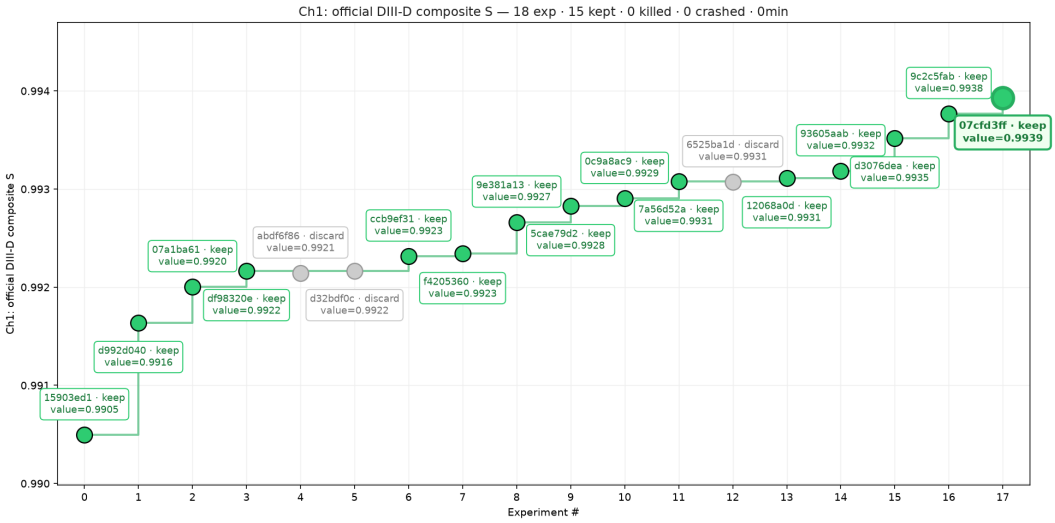


Figure 1: Official DIII-D S per scored DIII-D prediction file, with the running best. Grey points did not beat the best: temporal smoothing (`abdf6f86`), a MAST-only change resubmitting the `df98320e` DIII-D file (`d32bdf0c`), and the 27-member blend (`6525ba1d`).

Two model families account for most of the second half. The shape target lifted the official score by +0.00049 in two submissions (`f4205360` to `5cae79d2`); the grid model by +0.00081 in four (`12068a0d` to `07cfd3ff`), mostly through consistency (0.974978 $\rightarrow$ 0.978802). Adding members to an existing family stopped paying after the 9-model ensemble (13 and 17 members: 0.99277 and 0.99272 locally against 0.99277), and $R^2_{q_{95},\beta_N}$ has been 0.995946 in every submission since 4 October. In the 160-shot screens (Table 3) each of the four grid submissions was positive in all five 40-shot groups.

Table 3: Screens of the grid blends: paired $\Delta S \times 10^{-6}$ per 40-shot group against the previous official best, and pooled S on the 160 never-scored shots (`docs/experiments/fusioneq/screens.yaml`). The w75 candidate has not been submitted.

Blend	spare 1	spare 2	spare 3	valid.	scoring	S_{160}	Official
grid_v1 at 0.1	87	83	191	63	73	–	0.993186
grid_v2 at 0.5	441	502	2166	445	301	0.993167	0.993520
two grids at 0.6	288	340	1101	207	54	0.993643	0.993769
three grids at 0.65	171	150	557	200	102	0.993914	0.993927
grids at 0.75 (w75)	21	15	190	130	39	0.994000	not scored

On the 40 local scoring shots the official best blend scores $S = 0.994341$ (validation shots 0.993694) against 0.993927 officially; the local scoring shots have run high since the first entry (0.9915 locally against 0.9905 officially). Against the leaders (0.9964–0.9967) the gap is mostly consistency (0.9788 against about 0.988, weakest in κ and l_i), then the boundary (D_{LCFS} 0.0110 against 0.007) and q_{95}/β_N (0.9959 against 0.998; `docs/journey.md`). Projecting the true flux onto the base model's 192-component PCA scores 0.9988, so the representation is not the limit (`docs/diii-d-diagnosis.md`).

5 The funnel

Table 4 counts every stage, including runs that were killed or never scored.

Table 4: From trained runs to the submitted blend. Sources: `configs/schedules/`, the sweep writeups in `docs/experiments/fusioneq/mlp/`, `verdicts.yaml`, `screens.yaml`, `results/codabench.json`, and the git log of 11 October.

Stage	Count
MLP sweeps (v1–v21)	21 schedules, 115 scheduled iterations (trainings plus ensemble iterations). Not completed as scheduled: 3 of v2's (cut after 2 of 5), 2 of v4's (killed by a false-positive memory triage), 4 of v15's (killed by the GPU-hang triage, rerun as v15b).
Grid models	6 started: grid_v1, grid_v2, grid_v2_s1, grid_v3, grid_v2_gs, and grid_v4 (96 channels), which was lost to an OOM kill on 11 October and is retraining. grid_v3_ctx8 and grid_v3_env8 are being re-run.
Experiments with a verdict	27: 11 PROMOTE, 5 KEEP, 11 DISCARD.
Blend screens	12 candidates in `screens.yaml`, 6 of them submitted. On 11 October two blend builds were OOM-killed on the shared host (before `blend -no-score`), and every w75 build was OOM-killed (count not recorded) before the compact XGBoost loader.
Official submissions	35 rows in `codabench.json`: 3 candidates held back, 1 submitted without a separate score, 31 scored. They carry 17 distinct DIII-D prediction files; 15 set a new best (Table 2) and 2 did not.
Submitted blend	47 components: 15 base MLPs, 1 coefficient-tree model, 27 shape-target MLPs, 1 shape-tree model and 3 grid models.

6 Cross-machine transfer (MAST)

The learned models ingest DIII-D coils, so MAST gets a physics flux: its coil vacuum flux plus the measured plasma current spread as $J \propto (1-\rho^2)^{\alpha}$ over an ellipse placed where the Thomson density exceeds a fraction of its peak (`fusioneq/plasma.py`, `fusioneq/physics.py`). A global scale (-0.8724), a plasma-to-vacuum ratio (0.9005) and an offset (-0.0201) are fitted on DIII-D training

shots, where labels are allowed (`results/mast_physics.json`); the current entry uses $\alpha = 2.0$ and a threshold fraction of 0.4. q_{95} and β_N come from textbook formulas on the predicted boundary, the toroidal field and the Thomson pressure (volume-averaged β_N).

Table 5: Official MAST rows (`results/codabench.json`). R^2_ψ is after the scorer's global sign flip. `f4fa3fea` used terms fitted on the three labelled demo shots, which rule 3 forbids in the Final phase.

HF commit	MAST flux	S_{MAST}	R^2_ψ	$R^2_{q,\beta}$	D	Cons.	G
`15903ed1`	vacuum only	0	–	–	–	–	0
`737ce338`	calibrated physics flux	0.557707	0.873347	−0.259401	0.244564	0.009112	0.563058
`d32bdf0c`	α 2.0, frac 0.4, vol. β_N	0.604496	0.877339	0.320132	0.260607	0	0.609268
`f4fa3fea`	+ demo-fitted offset, lines	0.644578	0.935922	0.372570	0.260645	0	0.649666
`f2b36e3e`	label-free, half solenoid	0.540680	0.749632	0.320132	0.219799	0.011712	0.544070
`4c428222`	label-free, solenoid shape	**0.627752**	0.907922	0.320132	0.219674	0.011711	0.631688

The current entry is label-free: `4c428222` adds the shape part of halving the solenoid current (each frame's correction minus its grid mean, so the level is kept), which recovers about 58% of the banned offset's gain. With the `07cfd3ff` DIII-D file it gives $G = 0.631588$. The remaining R^2_ψ error is mostly a per-frame level: on the demo shots one global offset lifts R^2_ψ from 0.827 to 0.892 and the right per-frame offset to 0.957, and that offset tracks the solenoid current (correlation -0.99). A coil-geometry audit against published MAST geometry found no geometry error, so vessel currents are the suspect; nothing label-free calibrates them (`docs/mast-diagnosis.md`).

Disclosure: how α and the Thomson threshold were chosen. $\alpha = 2.0$ and the 0.4 threshold fraction were shortlisted by a 36-setting sweep scored on the three labelled MAST demo shots (demo S 0.537 $\rightarrow$ 0.584), which rule 3 now restricts to format inspection. No parameter is fitted to the demo shots and no demo-derived term enters the prediction. We keep the two values because official Development feedback, which rule 3 permits for comparing and selecting MAST pipelines, backs them: `d32bdf0c` ($\alpha = 2.0$, 0.4) scored MAST S 0.604496 against 0.6017 for `501b7986` (the previous setting), raising R^2_ψ 0.873 $\rightarrow$ 0.877 and $R^2_{q_{95},\beta_N}$ 0.294 $\rightarrow$ 0.320 while consistency fell 0.009 $\rightarrow$ 0 and D_{LCFS} rose 0.245 $\rightarrow$ 0.261. The q_{95} aspect-ratio cap of 0.65 was likewise compared officially against 0.7 (0.6017 against 0.6004). The organisers may read rule 3 more strictly; the label-free alternative is to re-select both values on official feedback alone.

7 Closed directions

Table 6: Do not retry (`verdicts.yaml`, `docs/mast-diagnosis.md`, `docs/journey.md`).

Direction	Measured result and reason
Grad-Shafranov loss on the grid model	At weight 50 the validation envelope loss rises to 0.0031 (grid_v2: 0.00075). On 160 shots it loses to the `07cfd3ff` blend (0.993914) as an extra grid (0.993827), as a replacement (0.993863) and alone (0.992890). Closed unless retried at weight 5 or less.
Free-boundary Grad-Shafranov for MAST shape	Checked on DIII-D, where labels are allowed: pinned centroid S 0.5035, consistency 0, R^2 of R_{axis} −9.2 and of κ −11.4; free centroid S 0.392. Without magnetics or vessel currents nothing holds the plasma shape.
Demo-fitted offsets on MAST	A solenoid offset and q_{95}/β_N lines fitted on the 3 labelled demo shots took MAST to 0.644578, but rule 3 forbids demo-shot fits in Final submissions.
Machine-agnostic grid model	Stopped at gate 0: DIII-D's plasma cannot be placed from Thomson the MAST way (R_0 correlation −0.16, a correlation 0.19 with EFIT), so the model could only learn a typical DIII-D plasma. A learned MAST flux correction had already lowered MAST S from 0.575 to 0.521.
More of one family	13 and 17 MLP members, wider (4096) or longer (240-epoch) members, the 160 spare shots as training data, whole-shot temporal models, full-resolution Thomson, history features: all flat or negative under G1.

8 What is not established

- **The w75 re-weighting.** Grids at 0.75 score 0.994000 on the 160 never-scored shots (+0.000086 over `07cfd3ff`, positive in all five groups), but it has not been scored officially, and the gain is below the size of past 160-to-official gaps.
- **Running grid variants.** grid_v3_ctx8 (± 8 frames of context) and grid_v3_env8 (heavier envelope weight) are being re-run, and grid_v4 (96 channels) is retraining after an OOM kill; none has a 160-shot result.
- **Group-level evidence is coarse.** Shot-level discordant counts are not recorded; with five paired 40-shot deltas a sign test reaches $p = 0.031$ at best.
- **MAST $R^2_\psi = 0.908$.** The per-frame level error that scales with the solenoid current is unexplained; vessel currents are the suspect and are untested. Our MAST consistency is 0.0117, while higher-ranked teams reach 0.24–0.89, and we do not know how. The demo-shot shortlist for α and the Thomson threshold is disclosed in Section 6; whether the organisers accept official feedback as the deciding evidence is theirs to rule.
- **The consistency gap.** 0.9788 against about 0.988 for the leaders, weakest in κ and l_i; its cause is not identified.
- **q_{95} / β_N.** Fixed at 0.995946 since 4 October. Direct heads and XGBoost on the scalars did not beat the base ensemble (+0.000055 validation, -0.000033 scoring), and how the leaders reach 0.998 is not known.
- **Final-phase reproduction.** Retraining the 47-component blend from the pinned commit within 24 A100-hours, and rebuilding the label-free MAST entry end to end, is untested. Per-member training times are logged in the sweep writeups (18–46 min for shape-target members), but the blend's total compute has not been summed.

9 Reproducibility

Code: `competitions/fusion-equilibrium/`, entry points in `fusioneq.cli` (`prepare`, `train`, `train-grid`, `blend`, `build`). Blend specs: `configs/blends/*.yaml`; sweep recipes: `configs/schedules/`. Locked versions (`uv.lock`): torch 2.14.0, xgboost 3.2.0, numpy 1.26.4, scikit-learn 1.9.1, autoresearch 0.38.0. Predictions are pushed to the private HF dataset `chaleong/fusion-eq-predictions`; the official best is HF commit `07cfd3fffbf78a0b012f384291e269d9b4318698`. Experiment card: `docs/experiment-card.md`. W&B project: `chaleong/fusion-equilibrium`.